

Pisier's Endpoint Class Need Not Be a p -Tensor: A Counterexample for Every $0 < p < 1$

Math discovery candidate full counterexample

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1 Status and source question

For $0 < p < 1$, Aleksandrov and Peller put

$$p_b = \frac{2p}{2-p}, \quad \mathcal{Y}_{p_b} = \ell^{p_b}(\ell^\infty) + (\ell^{p_b}(\ell^\infty))^t.$$

They ask whether this sufficient class of Schur multipliers is contained in their p -tensor product $\ell^\infty \otimes_p \ell^\infty$ [1]. The question appears in Remark 1 on page 20:

Remark 1. It is easy to see that $\ell^\infty \otimes_p \ell^\infty \not\subset \mathcal{Y}_{p_b}$ for $p \in (0, 1]$. Indeed, the infinite matrix $\mathbf{I}_\infty$ with entries identically equal to 1 belongs to $\ell^\infty \otimes_p \ell^\infty$ and does not belong to $\mathcal{Y}_{p_b}$. In the case $p \in (0, 1)$, the authors do not know whether $\mathcal{Y}_{p_b} \subset \ell^\infty \otimes_p \ell^\infty$ or not.

The answer is no for every $0 < p < 1$. In fact, the counterexample belongs to the smaller one-sided class $\ell^{p_b}(\ell^\infty)$.

2 Proof intuition

A flat n -by- n unitary has n equal singular values but entries of size $n^{-1/2}$. After multiplying it by $n^{-1/p_\sharp}$, where $p_\sharp = p/(1-p)$, the row-envelope cost is governed by

$$\frac{1}{p_b} = \frac{1}{p_\sharp} + \frac{1}{2}. \tag{1}$$

Thus a block with an additional scalar weight c costs c^{p_b} in the row class. On the other hand, its equal singular values force any p -tensor representation to spend essentially c^p coefficient mass in a window whose location is proportional to the block dimension.

Choose weights for which $\sum c_j^{p_b} < \infty$ but $\sum c_j^p = \infty$, and separate the dimensions so rapidly that these mandatory coefficient windows are disjoint. Sharing rank-one terms between blocks then cannot reduce the total cost.

3 The coefficient-window lemma

Recall that $A \in \ell^\infty \otimes_p \ell^\infty$ means that

$$A = \sum_{k \geq 1} x_k y_k^t, \quad \sum_{k \geq 1} \|x_k\|_\infty^p \|y_k\|_\infty^p < \infty. \tag{2}$$

Since $\ell^p \subset \ell^1$ for $p < 1$, such a series converges uniformly entrywise.

Lemma 1 (Scaled-unitary coefficient window). *Let $0 < p < 1$, put $r = p_\# = p/(1-p)$, and let U be an n -by- n unitary. Suppose $c > 0$ and*

$$cn^{-1/r}U = \sum_{k \geq 1} \lambda_k u_k v_k^t, \quad (3)$$

where $\|u_k\|_\infty, \|v_k\|_\infty \leq 1$ and $\lambda_1 \geq \lambda_2 \geq \dots \geq 0$ belongs to ℓ^p . For $0 < a < 1 < b$,

$$\sum_{an < k \leq bn} \lambda_k^p \geq c^p(1-a) - b^{p-1} \|\lambda\|_{\ell^p}^p. \quad (4)$$

Proof. Let $R = \sum_{k > an} \lambda_k u_k v_k^t$. What was removed from (3) has rank at most an . All n singular values of the left side are $cn^{-1/r}$, so the singular-value approximation inequality gives

$$\|R\|_{\mathcal{S}_p}^p \geq (1-a)n(cn^{-1/r})^p = c^p(1-a)n^p, \quad (5)$$

because $p/r = 1-p$.

For $T = \sum_{k > bn} \lambda_k u_k v_k^t$, the comparison of Schatten norms in dimension n , the rank-one trace norm estimate, and monotonicity of λ give

$$\begin{aligned} \|T\|_{\mathcal{S}_p} &\leq n^{1/p-1} \|T\|_{\mathcal{S}_1} \leq n^{1/p} \sum_{k > bn} \lambda_k \\ &\leq n^{1/p} (bn)^{-(1-p)/p} \|\lambda\|_{\ell^p} = b^{-1/r} n \|\lambda\|_{\ell^p}. \end{aligned} \quad (6)$$

Here the middle bound follows from $\sum_{k > bn} \lambda_k \leq \lambda_{[bn]+1}^{1-p} \sum_{k > bn} \lambda_k^p$.

Put $M = \sum_{an < k \leq bn} \lambda_k u_k v_k^t$, so $R = M + T$. The Schatten p -triangle inequality and (5)–(6) imply

$$\|M\|_{\mathcal{S}_p}^p \geq c^p(1-a)n^p - b^{p-1}n^p \|\lambda\|_{\ell^p}^p. \quad (7)$$

Each $u_k v_k^t$ has Schatten p -norm at most n , hence $\|M\|_{\mathcal{S}_p}^p \leq n^p \sum_{an < k \leq bn} \lambda_k^p$. Dividing (7) by n^p proves (4). $\square$

This is the source paper's identity-block window argument with the identity replaced by an arbitrary unitary; the proof shows that only the constant singular-value list matters.

4 Counterexample

Theorem 2. *For every $0 < p < 1$, with $q = p_b = 2p/(2-p)$,*

$$\ell^q(\ell^\infty) \not\subset \ell^\infty \otimes_p \ell^\infty.$$

Consequently $\mathcal{Y}_{p_b} \not\subset \ell^\infty \otimes_p \ell^\infty$.

Proof. Put $r = p_\# = p/(1-p)$ and define

$$c_j = (j+1)^{-1/p}, \quad b_j = c_j^{-2p/(1-p)} = (j+1)^{2/(1-p)}, \quad j \geq 1. \quad (8)$$

Choose positive integers n_j recursively so that

$$n_{j+1} \geq 4b_j n_j. \quad (9)$$

Let U_j be the normalized Fourier unitary of order n_j ; every entry of U_j has modulus $n_j^{-1/2}$. Partition $\mathbb{N}$ into successive sets E_j with $|E_j| = n_j$, and define the block diagonal matrix

$$W|_{E_j \times E_j} = c_j n_j^{-1/r} U_j, \quad W|_{E_j \times E_k} = 0 \quad (j \neq k). \quad (10)$$

By (1), the supremum of the absolute values in every row belonging to E_j is $c_j n_j^{-1/q}$. If α is the sequence of row suprema, then

$$\sum_i \alpha_i^q = \sum_{j \geq 1} n_j (c_j n_j^{-1/q})^q = \sum_{j \geq 1} c_j^q = \sum_{j \geq 1} (j+1)^{-q/p} < \infty, \quad (11)$$

since $q/p = 2/(2-p) > 1$. Thus $W \in \ell^q(\ell^\infty) \subset \mathcal{Y}_q$.

Suppose, toward a contradiction, that $W \in \ell^\infty \otimes_p \ell^\infty$. Normalize and reorder a representation (2) to obtain

$$W = \sum_{k \geq 1} \lambda_k u_k v_k^\dagger, \quad \|u_k\|_\infty, \|v_k\|_\infty \leq 1, \quad \lambda_1 \geq \lambda_2 \geq \dots \geq 0, \quad L := \|\lambda\|_{\ell^p} < \infty. \quad (12)$$

Compressing (12) to $E_j \times E_j$ does not increase either ℓ^∞ -norm. Lemma 1, with $a = 1/4$ and $b = b_j$, therefore gives

$$\begin{aligned} \sum_{n_j/4 < k \leq b_j n_j} \lambda_k^p &\geq \frac{3}{4} c_j^p - b_j^{p-1} L^p \\ &= \frac{3}{4} c_j^p - L^p c_j^{2p}. \end{aligned} \quad (13)$$

For all sufficiently large j , the last expression is at least $\frac{1}{2} c_j^p$. By (9), the intervals $(n_j/4, b_j n_j]$ are pairwise disjoint. Hence

$$\sum_{k \geq 1} \lambda_k^p \geq \frac{1}{2} \sum_{j \geq j_0} c_j^p = \frac{1}{2} \sum_{j \geq j_0} \frac{1}{j+1} = \infty, \quad (14)$$

contrary to (12). Thus $W \notin \ell^\infty \otimes_p \ell^\infty$. $\square$

5 Checks, scope, and novelty

The construction is genuinely endpoint: one flat block has unit-scale cost in both estimates, and the separation appears only after assembling infinitely many blocks with $\ell^q \setminus \ell^p$ weights. The error term in (13) is $L^p c_j^{2p} = o(c_j^p)$; its exponent is the reason for the particular choice of b_j . A bundled verification script checks (1), all exponent conversions, and the flat-unitary singular values for finite Fourier matrices.

The theorem answers only the inclusion asked in Remark 1. It does not settle automatic complete boundedness for $1 < p < \infty$, the spectral-measure extension problem, the diagonal Lorentz conjecture, or equality between the p -tensor product and its bounded pointwise closure.

Bounded exact-title, exact-phrase, arXiv-id, citation, and tensor-keyword searches through August 17, 2026 found no later answer. Later related works located concern triangular projections, Haagerup tensor products, smooth Schur multipliers, or analytic Schur multipliers. Novelty is a candidate claim pending specialist and database review.

References

- [1] A. B. Aleksandrov and V. V. Peller, *Schur multipliers of Schatten–von Neumann classes $\mathbf{S}_p$* , Journal of Functional Analysis 279 (2020), no. 8, 108683; arXiv:1910.06891.
- [2] A. Pietsch, *Operator Ideals*, North-Holland Mathematical Library 20, 1980.
- [3] G. Pisier, *Remarks on the non-commutative Khintchine inequalities for $0 < p < 2$* , Journal of Functional Analysis 256 (2009), 4128–4161.